

SANDHYA KANNEBOINA | DATA ENGINEER

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PROFESSIONAL SUMMARY

Data Engineer with 5+ years of experience designing, building, and optimizing scalable ETL/ELT pipelines, data warehouses, and data lake architectures across **AWS, Azure, and Hadoop** ecosystems. Skilled in **Python, SQL, Spark, Airflow, Databricks, dbt, Snowflake, Informatica, and SSIS** for automated, accurate, and governance-compliant data processing. Strong expertise in **data modeling, performance tuning, metadata management, and HIPAA/SOC2 compliance**. Experienced in creating **analytics-ready, secure, high-quality data infrastructure** to support **BI, ML, and Agile/DevOps initiatives**.

PROFESSIONAL EXPERIENCE

Data Engineer | BCBSA | Dallas, TX

June 2024 – Present

- Engineered **automated ETL/ELT pipelines** using AWS Glue, PySpark, and Python, processing high-volume healthcare data with a 35% latency reduction.
- Architected **data lake and warehouse frameworks** using S3, Redshift, Athena, and Snowflake ensuring schema evolution, lineage, and HIPAA compliance.
- Built **Spring Boot microservices** for real-time ingestion and REST/GraphQL API integration, enabling seamless downstream analytics access.
- Optimized AWS EMR clusters and EC2 configurations for distributed Spark workloads, improving throughput by 40%.
- Implemented **Kafka and Kinesis streaming** for near real-time data ingestion, achieving sub-second latency and 99.9% reliability.
- Developed **NoSQL data models** in DynamoDB and MongoDB to support unstructured data ingestion and high-velocity transactions.
- Designed SnapLogic pipelines integrating Salesforce APIs, REST endpoints, and healthcare systems, improving ingestion efficiency by 40%.
- Migrated **SSIS and Informatica ETL** workloads to AWS Glue and PySpark, reducing maintenance overhead by 30%.
- Built **metadata-driven data quality and governance frameworks** using AWS Lambda and Glue Catalog, enforcing DataOps standards.
- Implemented **data catalog and lineage tracking** using AWS Glue Data Catalog for 500+ datasets, improving data discoverability.
- Designed **RBAC-based access control** using IAM and KMS encryption ensuring HIPAA-compliant security for sensitive healthcare data.
- Orchestrated Airflow DAGs with SLA monitoring and alerting, achieving 99.99% data pipeline reliability.
- Tuned Snowflake and Redshift queries using dist/sort keys and caching for a 40% performance improvement.
- Built **Power BI dashboards** using Snowflake and Redshift for patient outcomes and KPI reporting.
- Monitored pipelines via CloudWatch and Datadog dashboards for anomaly detection and cost optimization.
- Delivered data modernization initiatives improving refresh speeds by 40% and enabling predictive analytics for care quality.

Data Engineer | Infosys | Hyderabad, India

January 2022 – January 2023

- Developed **Azure Data Factory (ADF) pipelines** and Databricks PySpark notebooks for transforming structured and unstructured data into Data Lake Gen2 and Synapse.
- Designed **star and snowflake schemas** in Synapse and SQL Server, improving query performance by 35%.
- Automated legacy ETL migration from SSIS to ADF pipelines, reducing manual workload by 50%.
- Implemented **dbt models with CI/CD** in Azure DevOps and Git, improving versioning and test automation.
- Created **Informatica Cloud and SnapLogic integrations** for Salesforce and REST APIs, enhancing interoperability.
- Built **data validation and reconciliation scripts** in Python/SQL ensuring accuracy across raw, curated, and semantic layers.
- Orchestrated ETL jobs using **Airflow DAGs** with dynamic dependencies and SLA monitoring.
- Enabled **Change Data Capture (CDC)** workflows using ADF and Delta Lake, automating incremental loads.
- Enhanced **pipeline resilience** through custom logging, retries, and Azure Monitor alerts.

- Implemented **role-based security (RBAC)** and AAD integration, ensuring ISO/SOC2 compliance.
- Developed **Power BI semantic layer** refresh automation for self-service analytics.
- Partnered with data scientists and BI teams to define SLAs, KPIs, and governance logic improving dashboard reliability.
- Delivered **Azure data platform modernization** increasing pipeline efficiency by 30% and traceability by 20%.
- Collaborated in Agile/Scrum teams for iterative delivery, backlog refinement, and sprint retrospectives.

Data Engineer | Wipro | Hyderabad, India

January 2019 – December 2021

- Developed **Hadoop-based ETL pipelines** using Hive, Sqoop, and Spark to process large-scale structured and unstructured data.
- Built **real-time streaming pipelines** using Kafka and Spark Streaming, enabling instant data processing for analytics dashboards.
- Migrated **on-prem Oracle and SQL Server** data to AWS S3 and Redshift ensuring schema validation and reconciliation.
- Automated **batch workflows** using Informatica and SSIS improving reliability and scheduling.
- Created **SnapLogic integrations** to bridge REST APIs, SaaS, and Hadoop data lakes improving data flow automation.
- Designed multi-zone **data lake architecture (raw → curated → consumption)** improving reusability and governance.
- Developed **Power BI dashboards** integrated with Redshift and Hadoop for executive-level metrics tracking.
- Orchestrated Spark and Hive jobs with Airflow for fault-tolerant scheduling and SLA adherence.
- Implemented **metadata-driven schema management** for dynamic job orchestration.
- Optimized Spark jobs using partitioning, broadcast joins, and caching, reducing runtime by 35%.
- Performed **data reconciliation and validation** using Python/SQL ensuring 100% accuracy.
- Managed deployments with Jenkins and GitHub including rollback strategies and approvals.
- Ensured **data governance and compliance** with HIPAA, GDPR, and SOC2 standards.
- Implemented **data retention and archival policies** in S3 and HDFS reducing costs by 20%.
- Delivered hybrid data platforms integrating ETL, streaming, and BI for faster decision-making.

EDUCATION & CERTIFICATIONS

Bachelors in Electronics and Communication Engineering– Chaitanya Institute of Technology and Science- CGPA:3.5/4

AWS Certified Solutions Architect – Associate | Microsoft Certified: Azure Fundamentals | Introduction to modern AI

TECHNICAL SKILLS

- **Cloud Platforms:** AWS (S3, Glue, Redshift, EMR, Lambda, Athena, EC2, Step Functions, CloudFormation), Azure (ADF, Data Lake Gen2, Synapse, Databricks, Fabric), Hadoop Ecosystem (Hive, HDFS, Sqoop, Yarn)
- **Big Data & ETL Tools:** Apache Spark, PySpark, Databricks, Airflow, dbt, Informatica, SnapLogic, SSIS, Spring Boot, Drools Rules Engine
- **Languages:** Python, SQL, Scala, Java, R, Shell
- **Data Modeling:** Star Schema, Snowflake Schema, Kimball, Data Vault, ERD, CDC (Change Data Capture)
- **Databases:** Snowflake, SQL Server, PostgreSQL, Oracle, MySQL, DynamoDB, MongoDB
- **Streaming & Messaging:** Kafka, Kinesis, SNS, SQS, Spark Streaming
- **Visualization & BI:** Power BI, Tableau, Qlik
- **CI/CD & DevOps:** Jenkins, GitHub Actions, Azure DevOps, Terraform, Docker, Kubernetes, Maven, Artifactory, uDeploy
- **Version Control:** Git, Bitbucket
- **Monitoring & Logging:** CloudWatch, Datadog, Splunk, Azure Monitor, Log Analytics
- **APIs & Integration:** REST APIs, Swagger, API Gateway, Salesforce APIs, SnapLogic
- **Security & Governance:** IAM, RBAC, KMS Encryption, VPC, SOC2, HIPAA, AAD Integration, DataOps, Data Governance
- **Methodologies:** Agile, Scrum, Kanban, SAsE, Continuous Improvement, Refactoring, Architecture Design, Code Quality (SonarQube), Performance Optimization